

Vibhu Kumar Singh

Bengaluru, Karnataka, India | +91 8076570505 | vibhukumar1224@gmail.com
linkedin.com/in/vibhu-kumar-singh | github.com/Vibhukumar10

PROFILE SUMMARY

Software Engineer with experience designing and building cloud-native backend systems and microservices using **Java (Spring Boot, Helidon)** and **RESTful APIs**. Proficient in developing scalable, high-performance applications, optimizing backend services, and automating deployments using **Kubernetes, FluxCD, Helm, Ansible, and Terraform**. Experienced in building and deploying applications on **Oracle Cloud Infrastructure (OCI)**, with a strong foundation in distributed systems, CI/CD, and cloud-native architectures. Passionate about developing reliable, maintainable backend services that deliver measurable business impact.

WORK EXPERIENCE

Software Engineer II

Oracle | Bengaluru, Karnataka, India

September 2024 - Present

- Designed and maintained Java based microservices with **Spring Boot** and **Helidon**, ensuring secure, reliable service communication and production readiness across distributed environments.
- Engineered and maintained **CNAP**, a cloud-native deployment automation platform using **FluxCD, Ansible, and Terraform**, eliminating manual configuration processes entirely.
- Built and deployed a **FluxCD MCP server** to support **GitOps workflows** by querying deployment status, debugging synchronization issues, detecting configuration drift, and sending notifications for failed or delayed deployments.
- Deployed, operated, and scaled backend microservices on **Kubernetes** with integrated **Prometheus** monitoring and centralized logging, improving system observability.
- Improved backend API performance by **30%** by refactoring request-processing flows, adding **connection pooling**, introducing **asynchronous execution**, and tuning database access through **Liquibase-managed indexes**; validated via **JUnit** and **integration tests**.
- Implemented **Python** based cross-repository watchdog health checks leveraging **OpenSearch** to aggregate signals, detect failures, and improve operational visibility across **CNAP** services.

Software Engineer

Oracle | Bengaluru, Karnataka, India

January 2023 - September 2024

- Built and maintained **RESTful backend services** using **Java (Helidon MicroProfile)**, developing **OpenAPI compliant APIs**, integrating **MySQL** for data persistence, and implementing **OAuth 2.0** for secure authentication.
- Built responsive front-end applications using **Oracle JET** and **React**, integrating seamlessly with **RESTful APIs** to deliver end-to-end features across the UI and backend services.

SKILLS

Programming Languages: Java, C++, Python, JavaScript, Shell

Frontend: HTML, CSS, React, Next.js, Oracle JET

Backend & Frameworks: Helidon, Spring Boot, Microservices, Apache Kafka, Node.js

Testing: JUnit, Mockito, Unit & Integration Testing, Test Automation

Cloud & DevOps: Git, Docker, Kubernetes, Terraform, Ansible, AWS, OCI

AI Tools & Technologies: LLM, RAG, MCP, Agentic AI, Copilot, ChatGPT

ACHIEVEMENTS

- Oracle Cloud Infrastructure Certified Foundations Associate
- Active **LeetCode** problem solver with strong focus on data structures, algorithms, and system design.

EDUCATION

Vellore Institute of Technology, Vellore | 2019 - 2023

Bachelor of Technology, Computer Science and Engineering | CGPA: 8.79